

Moving vehicle classification based on micro-Doppler signature

Yanbing Li*, Lan Du, Hongwei Liu
National Lab of Radar Signal Processing
Xidian University
Xi'an, China
*e-mail: xidianlyb@163.com

Abstract—Micro-Doppler information contained in the returned echo reflects the special structure of moving vehicles and can be utilized to discriminate wheeled vehicle and tracked vehicle. In this paper, generalized matched filter is employed to suppress clutter and a preprocessing is used to eliminate the influence of bulk velocity for Doppler spectrum firstly. Then features which depict the distinction between the wheeled and tracked vehicles are extracted and experimental results based on extracted features are presented.

Keywords—vehicle classification; micro-Doppler; adaptive clutter suppression; feature extraction.

I. INTRODUCTION

When a radar target moves, the relative velocity of the target along the line of sight (LOS) will induce frequency shift on the returned signal. If the target or any structure on the target has some motions in addition to its bulk translation, such as vibration or rotation, these motions will induce frequency modulation which generates sidebands frequencies around the Doppler frequency induced by target's bulk translation. This phenomenon is known as micro-Doppler effect [1]. Detailed mathematics analysis of micro-motion was presented in [2], and researches such as classification of human activities [3] and vehicles [4], [5] show that we can extract micro-motion information of interest from target's micro-Doppler signature, and use these micro-motion features to discriminate different targets.

For Doppler spectrum based moving vehicle classification, there are two facts that will directly affect the classification performance. One of the two is that ground clutter should be treated carefully. Unlike detection task, clutter suppression in classification should not only remove clutter from the returned signals, but also reserve the maximum information contained in the returned signals. In this sense, the classical clutter suppression technique i.e. Moving Target Indicator (MTI) may not suit for the classification task due to its nonlinear amplitude modulation. The other one is that the width of the Doppler spectrum will change with the bulk velocity. This fact may mislead classifier if this relationship between the width of the Doppler spectrum and bulk velocity is not handled properly. To deal with the effects of these two facts, we proposed a preprocessing method which is composed of two steps to eliminate undesired effects in this paper.

The paper is organized as follows. In section 2, we establish micro-motion model for wheel and track. Clutter suppression and Doppler spectrum normalization are discussed in section 3 and feature extraction method is presented in section 4. In the end, classification results are given in section 5 followed by the conclusion in section 6.

II. MICRO-MOTION MODEL OF VEHICLES

We assume the radial bulk velocity of vehicle is v , which has been compensated in the analysis of micro-motion structures of vehicles. For a moving wheel as shown in Fig. 1, which lies in the yo z plane, assume that there are N scattering points on the wheeled edge and the k th scattering point has an initial angle:

$$\theta_k = \theta_0 + 2\pi(k-1)/N \quad (1)$$

with $k = 1, 2, \dots, N$. The returned signal of wheel is represented as:

$$s(t) = \sum_{k=1}^N \frac{\xi_k}{\omega} \sum_{n=-\infty}^{\infty} e^{jn\theta_k} J_n(\beta) e^{jn\omega t} \quad (2)$$

where J_n denoting the n th-order Bessel function of the first kind, $\beta = 4\pi r_0/\lambda$, r_0 is the radius of rotation, λ is wavelength, ω is angle frequency of rotation, and ξ_k is the scattering coefficient of the k th scattering point.

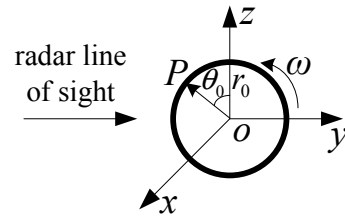


Figure 1. Micro-motion of wheel

For a moving track as shown in Fig. 2, the returned signal of the tracked is equal to the sum of returned signals generated

by parts AB , BC , CD , DE , EF and FA . The micro-motion of part AB or EF is rotation, and the returned signal of EF is added a phase relative to that of AB , which is caused by the length of track denoted by b . The micro-motion of part BC , CD , DE or FA is translation. Consequently, the returned signal of track is:

$$\begin{aligned}
 s(t) &= s_{AB}(t) + s_{BC}(t) + s_{CD}(t) + s_{DE}(t) + s_{EF}(t) + s_{FA}(t) \\
 &= \sum_{k=1}^{N_1} \frac{\xi_k}{\omega} \sum_{n=-\infty}^{\infty} e^{jn\theta_k^1} J_n(\beta) e^{jn\alpha t} + \\
 &\quad \sum_{k=1}^{N_2} \xi_k \exp[-j \frac{4\pi}{\lambda} (y_k^2 + vt \cos \varphi)] + \\
 &\quad \sum_{k=1}^{N_3} \xi_k \exp[-j \frac{4\pi}{\lambda} (y_k^3 + vt)] + \\
 &\quad \sum_{k=1}^{N_4} \xi_k \exp[-j \frac{4\pi}{\lambda} (y_k^4 + vt \cos \varphi)] + \\
 &\quad e^{j \frac{4\pi b}{\lambda}} \sum_{k=1}^{N_5} \frac{\xi_k}{\omega} \sum_{n=-\infty}^{\infty} e^{jn\theta_k^5} J_n(\beta) e^{jn\alpha t} + \\
 &\quad \sum_{k=1}^{N_6} \xi_k \exp[-j \frac{4\pi}{\lambda} (y_k^6 - vt)]
 \end{aligned} \tag{3}$$

where N_1, N_2, N_3, N_4, N_5 and N_6 denote the scattering point numbers of parts AB , BC , CD , DE , EF and FA , respectively. φ is the oblique angle of track, θ_k^q with $q=1,5$ denotes the initial angle of the k th scattering point from AB or EF respectively and y_k^q with $q=2,3,4,6$ denotes the initial location of the k th scattering point from BC , CD , DE or FA , respectively.

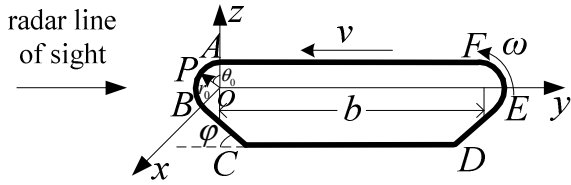


Figure 2. Micro-motion of track

Sample Doppler spectra for wheeled vehicle and tracked vehicle are shown in Fig. 3a and Fig. 3b respectively. The peak frequency in each spectrum represents the bulk-velocity, namely v , and the frequencies around 0 are induced by clutter. In practice, wheel is made of rubber which hardly scatters the electromagnetic wave. Consequently, it is difficult for radar to detect the micro-motion of wheel, and the Doppler spectrum of wheeled vehicle approximates a monophonic frequency which is induced by bulk-motion as in Fig. 3a. Unlike wheel, track is made of metal which is a well behaved scattering material for radar. Thus there is an abundance of micro-Doppler frequencies in the Doppler spectrum of track. As shown in Fig. 3b, the micro-Doppler frequencies between

frequency bins represent 0 and v are induced by the scattering points of BC , DE , partial AB , and partial FE which have the negative z-coordinate values. The micro-Doppler frequencies between v and $2v$ are induced by the scattering points of partial AB and partial FE which have the positive z-coordinate values. The micro-Doppler frequency located in $2v$ is induced by the scattering points of AF , which is not always observed in practice because of shelter.

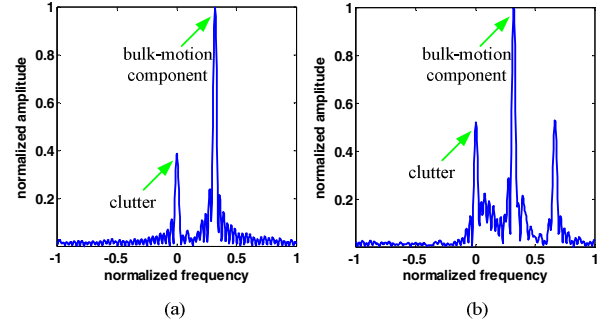


Figure 3. Doppler spectrum of vehicle. (a) Doppler spectrum of wheeled vehicle. (b) Doppler spectrum of tracked vehicle.

III. PREPROCESSING METHOD

A. Adaptive clutter suppression

For moving vehicles, there is clutter in the radar returned signal. This clutter occupies the narrow band frequencies around 0 in Doppler spectrum since it is usually immobile or has slow velocity. If the bulk velocity of vehicle is small, the bulk Doppler frequency will be close to 0. As a result, abundant micro-Doppler frequencies distributing between 0 and the bulk Doppler frequency are compacted in a narrow frequency band which is very close to 0. In this case, the purpose of clutter suppression is not only to remove clutter but also to preserve the micro-Doppler frequencies near 0, which is different from the goal of clutter suppression in target detection. Therefore, the clutter suppression for moving vehicles classification must be operated in a proper way, otherwise these micro-Doppler frequencies which are important to the following classification step might be suppressed or distorted undesirably.

If the terrain that surrounds the target has no upheaval in a certain range, the characteristic of ground clutter is approximately stable, and Doppler frequencies induced by clutter can be considered as colored noise. Thus the clutter suppression can be treated as signal detection in noise, and a technique to detect signal in colored noise, namely the generalized matched filter (GMF) [6], can be employed to deal with this situation. The Doppler spectrum of track vehicle after clutter suppression by generalized matched filter is shown in Fig. 4. The generalized matched filter achieves much better performance than moving target indicator (MTI) [7] especially for frequency band near 0. More specifics are preserved by generalized matched filter between 0 and bulk Doppler component. In a word, the generalized matched filter preserves

the most of Doppler frequencies of original signal except the clutter than MTI does, and the form of Doppler spectrum is similar to the original one.

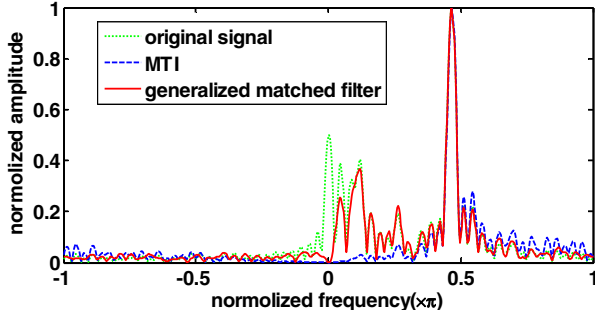


Figure 4. Clutter suppression by generalized matched filter

B. The velocity normalization of Doppler spectrum

According to the micro-motion model in section 2, the maximum velocity on the micro-motion structure is two times as large as the bulk velocity due to the structures of vehicles. Thus the width of the Doppler spectra of vehicles will change with different bulk velocities. As shown in Fig. 5, the Doppler spectra of vehicle which have high bulk velocity are wider than those have low bulk velocity.

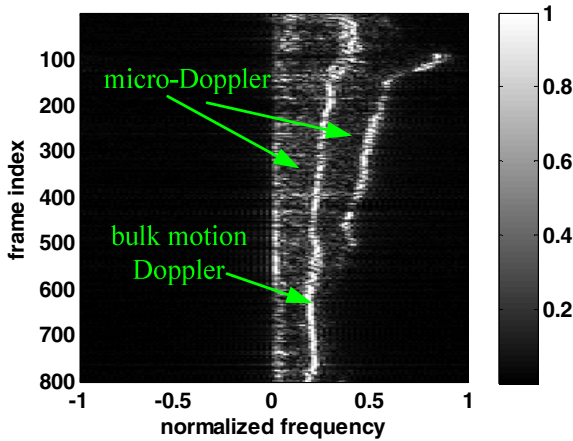


Figure 5. Group Doppler spectra of tracked vehicle

Because of the relationship between the width of Doppler spectrum and the bulk velocity, the characteristics of Doppler spectrum will be affected by bulk velocity. This is undesirable in the classification stage since we wish the classification results are robust to different bulk velocities. In order to eliminate the influence of bulk velocity, proper velocity normalization will be implemented. The normalization results are shown in Fig. 6. Doppler spectra before and after normalization are shown in the first and second row respectively. It can be seen that Doppler spectra with different bulk velocities have the same width after normalized by a reference velocity.

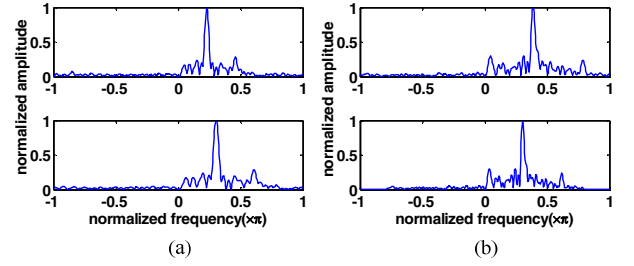


Figure 6. Velocity normalization. (a) Doppler spectrum with low bulk velocity. (b) Doppler spectrum with high bulk velocity.

IV. MICRO-DOPPLER FEATURE EXTRACTION

The distinction between the wheeled and the tracked is the number of harmonics contained in their returned signals. Based on the analysis in section 2, the spectrum of the wheeled is approximately a monophonic component besides clutter while the one of the tracked contains abundant frequencies. For a given Doppler spectrum namely $\mathcal{S} = \{S_i\}_{i=1}^L$, three features which describe the distinction between Doppler spectrum of the wheeled and the tracked are presented as follows:

The first feature is the entropy of Doppler spectrum:

$$\text{feature1} = -\sum_{i=1}^L p_i \ln p_i \quad (4)$$

$$\text{with } p_i = S_i / \sum_{i=1}^L S_i.$$

The second feature is the sum of the spectrum difference:

$$\text{feature2} = \text{sum}(\text{diff}(\mathcal{S})) \quad (5)$$

where $\text{sum}(\cdot)$ is the sum operation and $\text{diff}(\cdot)$ is the difference operation.

The third feature is the bulk-motion energy to micro-motion energy ratio:

$$\text{feature3} = b_energy(\mathcal{S}) / m_energy(\mathcal{S}) \quad (6)$$

where $b_energy(\cdot)$ is the operation to get the energy of bulk-motion component and $m_energy(\cdot)$ is the operation to obtain the energy of micro-motion components.

V. EXPERIMENTAL RESULTS

Measured data are collected from one wheel vehicle and one track vehicle under two experimental scenarios and only a single vehicle target is tested at one time. Firstly the vehicle recedes from radar, and secondly the vehicle approaches radar. We use the data under different target circumstances as training and testing set respectively to evaluate the generalization

performance of the extracted features, and employ Support Vector Machine (SVM) [8] as classifier for the features extracted by our method. Radial Basis Function (RBF) is selected as the kernel of SVM, and the parameter of RBF is optimized to have the minimum average classification error by searching in a certain range.

TABLE I. CLASSIFICATION RESULT

<i>Preprocessing method</i>	<i>The wheeled</i>	<i>The tracked</i>	<i>Average</i>
<i>MTI With normalization</i>	65.37%	61.68%	63.53%
<i>GMF Without normalization</i>	76.30%	78.51%	77.40%
<i>GMF With normalization</i>	84.77%	87.37%	86.07%

The classification results with different preprocessing methods are given in TABLE I. As we can see, the average classification accuracy for three preprocessing methods, i.e. MTI with velocity normalization, GMF without velocity normalization and GMF with velocity normalization, are 63.53%, 77.4% and 86.07% respectively. These results show that the GMF has much better performance than MTI to suppress clutter for the classification task since GMF can reserve the most of micro-Doppler information from the original signals. The results for GMF with and without velocity normalization show that the change of width of the Doppler spectrum, which is induced by the change of the bulk velocity, indeed influences the classification performance and it can be overcome by velocity normalization.

VI. CONCLUSIONS

The wheeled vehicle and the tracked vehicle have different micro-motion structures, which will induce micro-Doppler

modulation in the radar echo. This micro-Doppler information is promising resources for moving vehicle classification task. After proper clutter suppression and preprocessing for Doppler spectrum, features which describe micro-Doppler information can be extracted and satisfied classification results can be obtained.

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